

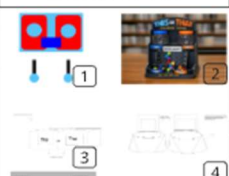
MyDesign Portfolio

Element A

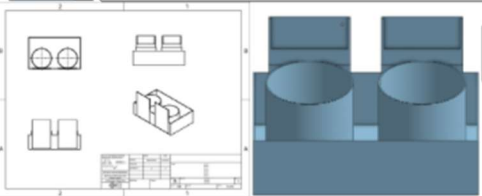
Furious Four

Problem definition

A local middle school librarian Mrs. Hybarger wants a this or that game to keep kids engaged while at the checkout desk. And to vote on things like which problem they want solved or what activities they'd rather do.

Initial Ideas	Decision matrix																																										
	<table><tr><th></th><th>Design 1 (Sketch 1)</th><th>Design 2 (Sketch 2)</th><th>Design 3 (Sketch 3)</th><th>Design 4 (Sketch 4)</th><th>Design 5 (Sketch 5)</th></tr><tr><td>Score</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td></tr><tr><td>Stability</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td></tr><tr><td>Value</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td></tr><tr><td>Cost</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td></tr><tr><td>Complexity</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td></tr><tr><td>Score</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td><td>0.2</td></tr></table>		Design 1 (Sketch 1)	Design 2 (Sketch 2)	Design 3 (Sketch 3)	Design 4 (Sketch 4)	Design 5 (Sketch 5)	Score	0.2	0.2	0.2	0.2	0.2	Stability	0.2	0.2	0.2	0.2	0.2	Value	0.2	0.2	0.2	0.2	0.2	Cost	0.2	0.2	0.2	0.2	0.2	Complexity	0.2	0.2	0.2	0.2	0.2	Score	0.2	0.2	0.2	0.2	0.2
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Prototype/Prototype Build



The original idea of the design is you have two voting boxes side by side that are designed similar to a basket ball hoop and the users will throw pompoms into the hoops to vote.

New Iterations

After careful review of all the feedback given from stakeholders and subject experts we made 3 main changes.

To hold the note cards we added on the backboard we designed it so that a notecard can be slotted into place .

To hold the pom pots before use we have extended the base to the max of 11x18 and then added sidewalls so that any unused pom pom or pom pom that was thrown and missed did not go on the ground.

We also moved each "hoop" farther apart so that accidental wrong votes will not happen and have them not diagonal not the board so they sit better on the table.

Problem justification

When Middle Schoolers are forced to go to the library they can often be disengaged or bored.

Built design photos



Testing plan

Test case	Test description	Test results	Test status
1 ex	The prototype must be able to hold the game board in a flat space.	Test case: N/A 1. 1x18x18 in 2. 1x18x18 in	Pass: mean of the trials is 3 kg Fail: mean of the trials is 2 kg
2 ex	The prototype must be able to hold the game board in a flat space.	Test case: N/A 1. 1x18x18 in 2. 1x18x18 in	Pass: mean of the trials is 3 kg Fail: mean of the trials is 2 kg
3	The prototype must be able to hold the game board in a flat space.	Test case: N/A 1. 1x18x18 in 2. 1x18x18 in	Pass: mean of the trials is 3 kg Fail: mean of the trials is 2 kg
4	The prototype must be able to hold the game board in a flat space.	Test case: N/A 1. 1x18x18 in 2. 1x18x18 in	Pass: mean of the trials is 3 kg Fail: mean of the trials is 2 kg
5	The prototype must be able to hold the game board in a flat space.	Test case: N/A 1. 1x18x18 in 2. 1x18x18 in	Pass: mean of the trials is 3 kg Fail: mean of the trials is 2 kg

Materials List

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
cardboard	cardboard	Crafts store	0.44	1	0.44
scissors		Crafts store	0.17	1	0.17
tape		Crafts store	0.11	1	0.11
glue		Crafts store	0.08	1	0.08



Feedback

Feedback	Comments	Response
1	The design is too complicated and hard to use.	We will simplify the design and make it easier to use.
2	The design is too expensive and hard to make.	We will use cheaper materials and make it easier to make.
3	The design is too small and hard to see.	We will make the design larger and easier to see.
4	The design is too noisy and hard to hear.	We will make the design quieter and easier to hear.
5	The design is too slow and hard to play.	We will make the design faster and easier to play.

Testing of the prototype: [Test.mov - Google Drive](#)

Prototype demonstration: [Demonstration.mov - Google Drive](#)

Design Requirements

The prototype must remain intact when held upside down. The prototype must fit in an 11 in. by 18 in. space. The prototype must be durable enough to remain sufficient intact even under vigorous use. The prototype has to be able to contain a large number of voting objects. The prototype has to be a single piece.

Problem Description and Elaboration

Studies show that people who are forced to do something will enjoy the activity less than a person who chooses to do the activity. (Find Citation) The same is with children. When kids are forced to go to the library during school-time due to school-curriculum, they enjoy much less than kids who like going to the library. In a survey posted by Gay Ivey and Karen Broaddus,(Gay Ivey and Karen Broaddus article link), it is mentioned that “Students valued independent reading...”, implying that children enjoy self-discovery. While kids are not enjoying their time in the library, it is easy for them to get frustrated and antsy during long waits.

Citations: Ivey, Gay, and Karen Broaddus. ““Just Plain Reading”: A Survey of What Makes Students Want to Read in Middle School Classrooms.” Reading Research Quarterly, vol. 36, no. 4, 12 Oct. 2001, pp. 350–377, <https://doi.org/10.1598/rrq.36.4.2>.

Stakeholder Concerns and Connections

The four primary stakeholders are Mrs. Hybarger (The Librarian), The students of Sky Ranch Middle-School, the parents of students, and other faculty at Sky Ranch Middle-School. We were not able to speak with the students or parents of students during this project, but they are still affected, therefore, they are stakeholders. The students have one of the most obvious reasons to be a stakeholder. Students are the one who will be most affected, due to the fact that they are the topic of the project. If we make a poor design, students will not benefit from it at all, making it pointless. Mrs. Hybarger is the other stakeholder who will be most affected by this project. If the team working on the device makes a sub-par device, Mrs. Hybarger will most likely get complaints and the activity that the device is supposed to correct will continue. The parents of students will also be affected. Students often talk about their day to their parents when they get home, so an upset in their day will transfer to the parents and make no one happy. Other faculty at the middle school will get affected a lot like Mrs. Hybarger and the parents of students. Students come complaining and upset, which causes emotional stress and frustration on the faculty's day. Frustrated students are also hard to work with, making this issue even more frustrating.

Problem importance

The idea of a bunch of antsy, upset, and frustrated middle-school aged children is daunting at best, terrifying at worst. When a large group of people are unhappy in a public place, it often makes the experience unpleasant for everyone. Groups of students that age are often influenced easily by peers, so if just a few students are acting-up, it is plausible that a large group would change their outward demeanor and attitude towards the library in general. The frustration and irritation snowballs into other people's lives, causing more problems that could be fixed at the start of the issue.